Data Storm 7.0: Modeling Unconstrained Latent Demand and Spatially-Aware Trade Spend Optimization

Team InsightAI - Final Technical Report

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Abstract

This document outlines the comprehensive technical methodology developed by Team InsightAI to estimate the unconstrained latent demand across a heterogeneous network of 80,000 FMCG retail outlets in Sri Lanka, and to distribute a LKR 5 Million promotional budget across the Western Province. Traditional forecasting models suffer from "bottleneck-learning," as they accurately predict what a shop will sell based on what it sold previously, failing to recognize that past sales were heavily constrained by supply chain inefficiencies, portfolio gaps, and stock-outs. Our approach, "Causal Spatio-Temporal Unconstraining," bypasses historical ceilings by implementing a parameterized Medallion Lakehouse cleaning pipeline, correcting portfolio imbalances via Regional Collaborative Filtering, mapping local catchment footfall drivers using vectorized spatial indexing with SciPy cKDTree, and training an asymmetric LightGBM Quantile Regressor ($p90$). Finally, we formulate a non-linear convex trade marketing optimizer solved via Lagrangian shadow price binary search to maximize incremental volume lift. This end-to-end framework bridges advanced spatial econometrics and practical business decision-support.

1 Data Forensics and Medallion Lakehouse Hygiene

The first stage of our pipeline focused on Data Engineering (DE) checks and "Data Exorcism" by identifying and neutralizing artifacts in the SFA (Sales Force Automation) data that contaminate demand signals. A core challenge in demand forecasting is that "Actual Sales" does not equal "True Demand." Sales data is merely the intersection of demand and supply.

1.1 System Anomalies: Ghosts and Negative Returns

To ensure the integrity of our pipeline, we built a comprehensive **Data Quality Framework** consisting of parameterized rules:

- **DQ_001 (Duplicate Detection):** Removing identical records uploaded due to server synchronization retries.
- **DQ_002 (Negative Returns Correction):** Adjusting net monthly sales where return volumes exceeded gross sales due to delayed logistics logging.
- **DQ_003 (Referential Integrity):** Resolving discrepancies between transactional records and the outlet master database.

Through this framework, we identified "System Ghosts"—extreme variance spikes caused by data upload failures or negative return adjustments that dropped net monthly sales to exactly zero. If left untreated, machine learning models naively learn these "logistical zeros" as "zero consumer demand." These records were quarantined and replaced with 24-month rolling medians. To validate our cleaning process, we utilized the **Coefficient of Variation (CV)**. Outlets with an unnaturally low CV over 24 months were flagged as "Censored," indicating their sales were hitting a hard systemic capacity ceiling rather than fluctuating organically.

plots/plot_1_ghost_exorcism.png

Figure 1: Variance Stabilization. Demonstrating the stabilization of our target variable after removing extreme system-generated noise and wholesale sell-in spikes.

1.2 Distributing Demand: Quarterly Smoothing

We observed that monthly sales data in the FMCG sector is heavily "lumpy" due to wholesale sell-in behaviors (e.g., loading up inventory before the Sinhala/Hindu New Year). To prevent the model from learning these artificial logistical spikes as actual consumer demand, we applied a **3-Month Quarterly Rolling Average**. This smoothing process distributes the demand "lumps" across the quarter, ensuring that the model learns the steady-state consumer appetite.

1.3 Handling the "Lazy Labour" Bias

A critical exogenous constraint on sales is representative behavior. We engineered a **Lazy Rep Flag** to identify months where specific routes saw irregular or low visit frequency. Instead of treating these as "low demand" periods, our model flags them as "representative-constrained." We used the maximum flag per month to identify if a demand signal was authentic or merely a byproduct of secondary sales execution failure.

1.4 Dynamic Tiering: Correcting Static Labels

We observed that the static `Tier` labels in the provided outlet master data were stale and did not reflect current business realities. To provide our clustering algorithms with accurate shop sizes, we implemented **Dynamic Tiering**. We recalculated each outlet's tier classification based on its 24-month rolling average volume, ensuring our peer groups were built on current ground truth rather than outdated metadata.

2 Portfolio Correction via Regional Collaborative Filtering

Many shops underperform not because local demand is low, but because they lack the core products their local peers successfully sell. We implemented a **Collaborative Filtering** logic where outlets were grouped into "Regional Peer Groups" based on their Distributor ID, Outlet Type, and Dynamic Tier. For each group, we identified "Core SKUs" which are products sold by more than 50% of the peers. If a shop was missing a Core SKU, we synthetically imputed the median peer volume for that product. This explicitly "unconstrains" the outlet's potential by recognizing that if a specific beverage sells in 70% of local eateries, the 30% missing it have an untapped latent demand for it.

3 High-Performance Spatial Enrichment Engine (cKDTree V2)

To provide the model with physical context, we extracted high-resolution spatial data. In enterprise environments (such as Uber or DoorDash), calculating the exact distance between 20,000 retail outlets and 100,000 Points of Interest (POIs) creates an $O(N \times M)$ "N-Squared Problem." A naive cross-join requires 2 billion distance calculations, taking hours and crashing local servers.

3.1 Vectorized Spatial Indexing with SciPy cKDTree

To achieve performance suitable for production, we implemented a vectorized spatial engine using **SciPy cKDTree**. By projecting Latitude/Longitude (EPSG:4326) coordinates into a metric system (UTM Zone 44N / EPSG:5234) where 1 unit corresponds to 1 meter, we indexed the POI coordinate space. This allowed us to query nearest neighbors within a specific metric radius (e.g., 500m catchment) in **1.75 seconds**, a 90% reduction in compute overhead compared to legacy flat-buffer spatial joins.

3.2 Continuous Distance-Decay and Huff Gravity Models

Standard spatial buffers treat points 10 meters away and 490 meters away as identical. We replaced this with non-linear models:

- **Continuous Exponential Distance-Decay:** For catchment drivers, the influence decays exponentially:

$$D_i = \sum_j e^{-\lambda d_{ij}}$$

  where $\lambda = 0.003$ (halving influence at approximately 230m).

- **Competitive Catchment Saturation (Huff Gravity):** For competitors, the competitive pressure is modeled using the inverse-square law:

$$C_i = \sum_j \frac{1}{(d_{ij} + 1)^\beta}$$

  where $\beta = 2.0$ reflects high local friction and demand cannibalization.

3.3 Composite Features: Latent Opportunity Ratio

Using these continuous fields, we constructed the **Latent Opportunity Ratio**:

$$\text{Latent\_Opportunity\_Ratio}_i = \frac{\text{total\_driver\_gravity}_i}{\text{competitive\_saturation\_index}_i + \epsilon}$$

Outlets with high driver gravity and extremely low competitor density are flagged as **is_isolated_goldmine**. The model identified 922 such outlets across the network.

plots/plot_2_spatial_hotspots.png

Figure 2: Spatial Catchment Hotspots. Visual proof of hidden high-density demand nodes in the Western Province mapped via Overture Data.

4 Causal Base Logic and Peer Grouping

Our statistical approach addresses the "Right-Censored" nature of FMCG sales data. We assume that Observed Sales = min(Demand, Supply Capacity). To solve this, we must estimate the uncapped potential.

4.1 Statistical Peer Grouping: K-Means & Elbow Method

Before predicting potential, the model must understand "Peer Behavior." We used K-Means clustering to create behavioral cohorts based on static spatial features. This allows the model to compare a "Censored" shop (one hitting a ceiling) to its "Star" peers in the same spatial cluster. We used the **Elbow Method** to validate the overall clustering structure, but deliberately over-indexed to $k = 30$ clusters. Rather than creating broad regional groups, selecting $k = 30$ ensures our clusters capture hyper-local "Micro-Segments." This guarantees that each peer group is tightly bound by highly specific behavioral and spatial homogeneity.

`plots/plot_5_elbow_method.png`

Figure 3: The Elbow Method. Statistically validating our peer-group selection strategy.

4.2 Methodology: LightGBM ($p90$) and Validation Strategy

Standard regression algorithms (like OLS) predict the *mean*, which in our case is a constrained historical average. To estimate the true ceiling, we optimized a **LightGBM Quantile Regressor** using the **Pinball Loss** function with $\alpha = 0.90$. This asymmetric loss function forces the algorithm to ignore median constraints and aggressively map the 90th percentile "Star Potential" of a shop based on its spatial catchment and engineered interaction terms (like `Tuition_Weekend_Gravity`).

Validation on Natural Demand: To ensure the model learned true potential rather than constrained behavior, we employed a strict **Chronological Train/Val/Test Split (80/10/10)**. Crucially, the model was trained *exclusively* on data from "Uncensored" shops (outlets that successfully navigate supply constraints). By isolating this pure demand signal, the model learns the true causal mapping between spatial drivers and volume, which it subsequently applies to the "Censored" shops to accurately predict their unconstrained potential.

plots/plot_3_feature_importance.png

Figure 4: LightGBM Information Gain. Proof that the model relies on causal spatial logic rather than simply memorizing historical sales.

4.3 Validating the Uncapped Ceiling: Structural Safety Floor

To ensure business logic consistency and prevent algorithmic hallucination, we enforced a **Structural Safety Floor**:

$$\text{Final Prediction} = \max(\text{Historical Max Volume}, \text{Model p90 Output})$$

This logic ensures that the prediction is always grounded in an outlet's proven historical reality. The model will never predict a potential lower than what a shop has already achieved, but it is mathematically allowed to "break through" historical ceilings if the Overture spatial indicators and peer-group performance justify a higher potential.

plots/plot_4_scatter_unconstrained.png

Figure 5: Breaking the Demand Ceiling. Actual sales (blue) hit systemic capacity ceilings, while our 90th-percentile architecture maps the latent potential (green stars) hiding above the constraint.

5 Spatially-Aware Marketing Spend Optimization

To support C-suite resource allocation, we formulated an optimization framework to distribute a LKR 5 Million promotional budget across the Western Province for January 2026. The objective is to maximize the additional sales volume gained compared to normal historical baseline sales patterns.

5.1 Mathematical Model Formulation

For each outlet i in the Western Province:

- P_i: Predicted potential volume for January 2026 (Liters).
- H_i: Historical average monthly volume baseline (Liters).
- $G_i = \max(0, P_i - H_i)$: Growth Gap, representing the maximum latent demand we can unlock.
- S_i: Allocated promotional spend in LKR.

We model the incremental sales lift $L_i(S_i)$ using a concave exponential response curve (diminishing returns):

$$L_i(S_i) = G_i \left(1 - e^{-\alpha_i S_i}\right)$$

where α_i is the spend responsiveness coefficient of outlet i. To ensure spatial awareness, we define:

$$\alpha_i = \alpha_{\text{base}} \times (1 + 2 \times \text{scaled\_ratio}_i) \times (1 + 0.5 \times \text{Cooler\_Count}_i)$$

where $\alpha_{\text{base}} = 0.00005$, Cooler_Count_i represents the number of coolers currently deployed, and scaled_ratio_i is the log-transformed and normalized `latent_opportunity_ratio`. This formulation ensures that spatial "goldmines" and assets like coolers boost the marginal ROI of promotional spend.

The optimization problem is:

$$\max_S \sum_i G_i \left(1 - e^{-\alpha_i S_i}\right)$$

subject to:

$$\sum_i S_i \leq 5,000,000 \quad \text{and} \quad 0 \leq S_i \leq 50,000 \quad \forall i$$

The upper bound of LKR 50,000 per outlet represents a physical operational limit.

5.2 Lagrangian Shadow Price Solver

Because the objective function is concave and separable, we solve it globally using the method of Lagrange multipliers. The Lagrangian is:

$$\mathcal{L}(S, \lambda) = \sum_i G_i \left(1 - e^{-\alpha_i S_i}\right) - \lambda \left(\sum_i S_i - B\right)$$

Solving $\frac{\partial \mathcal{L}}{\partial S_i} = 0$ yields the optimal spend as a function of the shadow price λ:

$$S_i(\lambda) = \min\left(S_{\max}, \max\left(0, \frac{1}{\alpha_i} \ln\left(\frac{G_i \alpha_i}{\lambda}\right)\right)\right)$$

Since $\sum S_i(\lambda)$ is strictly decreasing in λ, we perform a binary search to find the optimal shadow price λ^* that satisfies the budget constraint. This solver guarantees convergence in $O(N \log(1/\epsilon))$ time (milliseconds), bypassing the convergence risks and computational limits of generic non-linear programming tools.

5.3 Optimization Results

The optimizer was applied to the 9,000 Western Province outlets, allocating LKR 4,999,999.90. A total of 1,378 high-ROI outlets were funded, unlocking an estimated **363,227.12 Liters** of incremental demand. The average spend for funded outlets was LKR 3,628.45.

Table 1: Trade Spend Allocation Summary by Distributor

Distributor ID	Outlets	Funded Outlets	Total Spend (LKR)	Incremental Lift (L)
DIST_W_01	3,020	447	1,745,155.13	127,630.67
DIST_W_02	2,989	461	1,668,049.55	120,460.58
DIST_W_03	2,991	470	1,586,795.22	115,135.87

6 Outlet Intelligence Web App and GenAI Explainability

To make these complex spatial and statistical insights accessible to business users, we developed a local **Streamlit Web Application** (`app.py`). The dashboard features:

- **Executive KPI cards:** Dynamic display of network potential, baselines, opportunity gaps, and budget allocations.
- **Interactive Plotly Maps:** Visualizing outlet geographical coordinates color-coded by Dynamic Tier, with bubble size proportional to January 2026 potential.
- **Live Budget Recalculator:** Allows executives to adjust the Western Province budget slider (LKR 1M to 10M) and trigger the Lagrangian solver in real time.
- **Outlet Drilldown Search:** Enables searching specific Outlet IDs to view metadata, cooler count, and spatial gauges.

6.1 Hybrid Explainable AI (XAI) Module

The application integrates an LLM-based component (`src/gold/llm_xai.py`) to generate clear business explanations for outlet scores. The module uses a hybrid architecture:

- **Gemini API Integration:** If `GEMINI_API_KEY` is present, it formats a detailed prompt describing the outlet's potential, growth gap, dynamic tier, cooler count, and spatial gravity signals, requesting a 3-sentence executive summary.
- **Local Fallback Engine:** If offline or no key is present, a deterministic rule-based template engine generates a highly-contextual narrative to ensure zero downtime.

7 GenAI Transparency Log

Team InsightAI utilized Generative AI strictly as an engineering accelerator to increase development velocity during the final round.

Table 2: InsightAI GenAI Usage and Prompt Transparency

Workflow Stage	LLM Application	Engineering Rationale / Prompt Examples
Spatial V2 Engine	SciPy cKDTree optimization	"Write a vectorized query using scipy.spatial.cKDTree to find nearest POIs within 500m metric UTM coordinates and calculate continuous exponential decay."
Spend Optimization	Lagrangian Binary Search	"How do I implement binary search on lambda for $S_i = \max(0, 1/\alpha_i \ln(G_i\alpha_i/\lambda))$ subject to budget B in numpy?"
Explainable AI (XAI)	System prompt engineering	"Translate these retail metrics [potential, history, coolers, spatial gravity] into 3 business sentences explaining why this shop got its score and recommend action."
Aesthetic Dashboard	Streamlit layout & CSS injection	Generative guidance on creating modern, dark-slate UI containers and glassmorphism elements using custom CSS.

8 Conclusion and Impact Summary

Team InsightAI has successfully delivered a production-ready spatial analytics and budget optimization pipeline. By resolving the N-Squared spatial indexing bottleneck with SciPy cKDTree and modeling competitive saturation with Huff gravity, we extracted a reliable latent opportunity signal. Quantile regression ($p90$) enabled demand unconstraining by modeling peak performance. The Lagrangian optimizer guarantees that the promotional budget is allocated to outlets with the highest spatial opportunity and cooler density, maximizing incremental volume. The Streamlit dashboard bridges these algorithms with intuitive interfaces and LLM explanations, establishing a robust framework for trade-spend decision-making.